

Sean Kim

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EDUCATION

University of Waterloo

Waterloo, Ontario

Honours Bachelor of Mathematics in Statistics, minor in Computer Science and Economics (CGPA: 3.94)

Expected 2029

- President's Scholar of Distinction; Term Distinction all applicable terms

EXPERIENCE

AXL Labs

Toronto, Ontario

AI Engineer

Sep 2026

- Develop MVPs for portfolio companies, integrating LLM and ML models into full-stack applications from prototype to release

RBC Insurance

Toronto, Ontario

Investment Analyst, CIO Office

Jun 2026 – Aug 2026

- Authored annual reviews of \$1.1B of **real estate and infrastructure equity** fund performance and co-built the diligence framework screening 20+ managers for \$1B of **CLO, IG/HY infrastructure debt, and infrastructure equity** mandates.

RBC Insurance

Toronto, Ontario

Actuarial Analyst, CIO Office

Jan 2026 – Apr 2026

- Modeled liability cash flows, key rate durations, and **convexity** (Pandas, OpenPyXL) to assess yield curve sensitivity and strengthen **duration matching** within the ALM team.
- Automated recurring ALM workflows via Python using **Win32Com** and **BLPAPI**, reducing manual overhead.
- Analyzed **infrastructure equity and debt** investment opportunities to support portfolio manager decisions.

Korn Ferry International

Seoul, South Korea

Consulting Intern, Organizational Consulting

Jun 2025 – Aug 2025

- Conducted HR due-diligence for the cross-border acquisition of a content startup valued at **\$500M**.

Aleph Lab (YC F25, formerly Lucid AI)

Seoul, South Korea

Co-Founder

Mar 2025 – Jun 2025

- Built a MacOS local AI agent in Swift and Python using the Gemini API (**Valuation: \$1M**); led sales and investment pitches.

2nd Artillery Brigade, Republic of Korea Army

Hwacheon, South Korea

Fire Direction Squad Leader, Intelligence & Operations

Sep 2023 – Mar 2025

- Managed **10+enlisted soldiers** in battalion HQ's Fire Direction Center; awarded **Brigadier General Commendation** twice.
- Reduced subordinate unit response times by **40%** via an **Excel VBA dashboard** digitizing real-time artillery status.

Hanwha Life Insurance Co.

Seoul, South Korea

Summer Analyst, Private Equity

Jun 2023 – Aug 2023

- Underwrote PE deals up to **\$100M**, conducting financial modeling and analysis of company filings and sell-side reports.

PROJECTS

Pairs Trading Engine with Kalman Filter | *Python, NumPy, statsmodels, Pandas*

- Screened cointegrated equity pairs via **Engle-Granger**, ADF, and Benjamini-Hochberg correction; implemented a **Kalman filter** to track a time-varying hedge ratio β_t , eliminating OLS errors-in-variables bias
- Backtested 15+ years with causal z-score signals and 5 bps costs; achieved out-of-sample **Sharpe 1.1** and max drawdown <12%

Market Regime Detection & Strategy Switching | *Python, hmmlearn, NumPy, Pandas, matplotlib*

- Built a **Gaussian HMM** from scratch (Baum-Welch EM, Viterbi) on volatility, VIX, and skewness features; selected regime count via **BIC** and validated against 2008 GFC, 2020 COVID, and 2022 rate shock
- Applied causal forward-only filtering and **Kelly-criterion** sizing per regime, reducing drawdown in high-volatility states

TripScape | *Python, TypeScript, SQL, React, Tailwind, FastAPI, PostgreSQL, SQLAlchemy, Docker*

- Architected and deployed a full-stack collaborative platform (React & TypeScript frontend, async FastAPI & PostgreSQL backend) engineering a unified state-management layer for real-time synchronization across Kanban, calendar, and map views.
- Implemented **real-time multi-user editing** via per-trip WebSocket broadcasts originating in the service layer, with client-side auto-reconnect and self-echo suppression to keep optimistic UIs flicker-free.
- Shipped on **Google Cloud Run, Vercel, and Neon Postgres**, deployed CI/CD pipeline with **Github Actions**.

NewsChecker | *Python, PyTorch, Huggingface Transformers, Pandas, Scikit-learn, NumPy, OpenAI API*

- **Fine-tuned** DeBERTa-v3-base on 3,700 expert-annotated news sentences for political bias detection, achieving macro F1 0.870 $\pm$ 0.005 and ROC AUC 0.937 across 3 random seeds with bootstrap 95% confidence intervals.
- Designed **outlet-holdout and topic-holdout ablations** with in-distribution controls to isolate OOD generalization gaps.
- Cut classification error $\sim$ **41%** versus a GPT-4o-mini zero-shot baseline (macro F1 0.87 vs 0.78) on a seeded 200-sentence holdout; threshold tuning raised biased-class F1 to 0.92.

SKILLS & INTERESTS

Languages: Python, SQL, Swift, JavaScript, TypeScript, VBA (**Human Languages:** English, Korean, French)

Frameworks & Tools: React, FastAPI, SQLAlchemy, PostgreSQL, Docker, GitHub Actions, OpenAI API, Gemini API

Libraries: NumPy, Pandas, statsmodels, matplotlib, hmmlearn, PyTorch, Hugging Face Transformers, Scikit-learn, BLPAPI

Interests: Defense industry, Robotics, Public Policy, Formula 1, Poker, Golf, Bartending, Drumming